PROJECT 1

Evolutionary Computation for Emerging AI

Individual Research Project · 40% of Final Course Grade

Course	DSAI6707 Advanced Computational Intelligence
Project Type	Individual
Submission Deadline	15 November 2026

1. Project Overview

This project asks you to connect the evolutionary computation techniques studied in this course (e.g., Genetic Algorithms, Particle Swarm Optimization, Differential Evolution, and Multi-objective Evolutionary Optimization) with emerging directions in artificial intelligence. **You should select a frontier topic, understand the state of the art through a focused literature review, identify a meaningful research opportunity, and develop and experimentally evaluate an innovative solution.** The interaction may go in either direction: evolutionary computation may be used to enhance emerging AI technologies, or modern AI techniques may be used to improve evolutionary algorithms. Hybrid approaches are also welcome.

2. Learning and Research Objectives

The project is intended to move beyond implementing a standard evolutionary algorithm on a conventional benchmark. By completing the project, you should demonstrate the ability to:

- understand and critically analyze recent research at the intersection of evolutionary computation and emerging AI;
- formulate a clear and meaningful research problem;
- design or adapt computational intelligence techniques to address that problem;
- introduce a reasonable degree of innovation based on existing work; and
- conduct rigorous comparative experiments and draw evidence-based conclusions.

3. Suggested Research Directions

The following topics are examples for inspiration rather than an exhaustive list. You are encouraged to explore other frontier directions when they have a clear connection to evolutionary computation.

- Evolutionary LLM architecture search
- Evolutionary prompt engineering
- Automatic algorithm design, discovery, or configuration
- LLM-assisted evolutionary computation
- Transformer-based or learned optimizers
- Evolutionary multi-agent systems
- Evolutionary fine-tuning and model adaptation & merging
- Self-evolving AI systems
- Multi-objective optimization for modern AI systems
- Evolutionary optimization of agents or world models

4. Expected Research Depth

This is an open-ended research project. There is no single correct solution. A strong project should first establish what has already been achieved in the selected area and then build on that understanding to make a meaningful technical contribution.

Simply applying an off-the-shelf GA/PSO/DE implementation to a new dataset or task is generally insufficient for a high mark. Innovation may arise from, for example, problem formulation, representation, fitness/objective design,

search operators, selection mechanisms, multi-objective formulation, interaction with an LLM/agent/model, or a new integration of existing techniques.

5. Literature Review Requirements

You must conduct a focused review of recent literature to establish the state of the art, identify limitations or gaps, and motivate your proposed work.

- Review at least 20 relevant papers published within the most recent three years (2024–2026).
- The review should synthesize the literature by themes, methods, trends, strengths, limitations, or research gaps. It should not be a collection of paper-by-paper summaries.
- The literature review should directly support the motivation and design of your proposed method.

6. Experimental Requirements

Experimental comparison is mandatory. Your experiments should provide convincing evidence about whether and why the proposed approach works.

- Include appropriate baselines, including relevant existing methods and/or conventional evolutionary approaches where applicable.
- Use meaningful quantitative evaluation metrics that match the research problem.
- For stochastic algorithms, conduct multiple independent runs where computationally feasible and report appropriate statistics (e.g., mean and standard deviation).
- Experimental settings, datasets/tasks, model configurations, hyperparameters, and evaluation procedures should be described clearly enough to support reproducibility.

7. Required Deliverables

Submit all of the following by the deadline **15 November 2026**:

1. **Project Report (PDF):** A self-contained research report covering the problem and motivation, literature review, methodology, experimental setup, results and analysis, discussion, conclusion, and references.
2. **Source Code:** Complete implementation used in the project, together with a README explaining environment setup, dependencies, data preparation, execution, and evaluation.
3. **Experimental Data and Results:** Raw or processed results sufficient to support the findings reported in the project, including relevant logs, result files, or evaluation outputs.
4. **Supplementary Materials (if applicable):** Additional configurations, prompts, visualizations, videos, model outputs, or other materials that help demonstrate or reproduce the work.

8. Assessment

The project contributes 40% of the final course grade. Assessment will consider both the quality of the research idea and the quality of its execution. In particular, projects will be judged with reference to the following dimensions:

Assessment Dimension	What Will Be Considered
Research Problem & Motivation	Significance, clarity, relevance, and understanding of the selected frontier topic.
Literature Review	Coverage of recent work, critical synthesis, and identification of meaningful research gaps.
Innovation & Technical Design	Originality, technical depth, appropriateness of the EC–AI integration, and quality of the proposed method.
Experimental Evaluation	Quality of baselines, rigor of comparisons, completeness of experiments, and depth of analysis.
Overall Completion & Research Quality	How completely and convincingly the proposed research idea is implemented and validated.
Report, Code & Reproducibility	Clarity of presentation, quality of implementation and documentation, and reproducibility of the reported results.

Important: Innovation and completion are central to the final assessment.

9. Reference Papers

The following papers illustrate representative research directions at the intersection of evolutionary computation and emerging AI. They are provided as starting points rather than an exhaustive reading list. You are strongly encouraged to explore recent literature beyond these examples and identify your own research direction.

[1] X. Wu, S.-H. Wu, J. Wu, L. Feng, and K. C. Tan, "Evolutionary computation in the era of large language model: Survey and roadmap," IEEE Transactions on Evolutionary Computation, vol. 29, no. 2, pp. 534–554, 2025.

[2] C. Fernando, D. S. Banarse, H. Michalewski, S. Osindero, and T. Rocktäschel, "Promptbreeder: Self-referential self-improvement via prompt evolution," in Proceedings of the 41st International Conference on Machine Learning (ICML), 2024, pp. 13481–13544.

[3] R. T. Lange, Y. Tian, and Y. Tang, "Large language models as evolution strategies," arXiv preprint arXiv:2402.18381, 2024.

[4] X. Wu, Y. Zhou, and K. C. Tan, "Building LLMs like LEGO: Two-dimensional architecture reassembly of large language models," in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL), 2026, pp. 44939–44956.

[5] T. Akiba, M. Shing, Y. Tang, Q. Sun, and D. Ha, "Evolutionary optimization of model merging recipes," Nature Machine Intelligence, vol. 7, pp. 195–204, 2025.

[6] H. Qiu, X. Wu, L. Feng, and K. C. Tan, "Fine-grained model merging via modular expert recombination," arXiv preprint arXiv:2602.06552, 2026.

[7] Y. Wang, C. Peng, X. Wu, S.-H. Wu, and Z.-H. Zhan, "How to guide LLM generation: Dual-surrogate guided search for automated heuristic design," arXiv preprint arXiv:2607.13911, 2026.

[8] F. Liu et al., "A systematic survey on large language models for algorithm design," ACM Computing Surveys, vol. 58, no. 8, Art. no. 218, pp. 1–32, 2026.

[9] Y. J. Ma, W. Liang, G. Wang, D.-A. Huang, O. Bastani, D. Jayaraman, Y. Zhu, L. Fan, and A. Anandkumar, "Eureka: Human-level reward design via coding large language models," in International Conference on Learning Representations (ICLR), 2024.

[10] S. Hu, C. Lu, and J. Clune, "Automated design of agentic systems," in International Conference on Learning Representations (ICLR), 2025.

[11] S. S. Li, R. Xin, Y. Wang, T. Xiao, R. Shao, Z. Hao, M. Sclar, F. Brahman, P. W. Koh, and Y. Tsvetkov, "EvoLM: Self-evolving language models through co-evolved discriminative rubrics," in International Conference on Learning Representations (ICLR), 2026.